

New Constructions of Antimagic Graph Labeling

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Abstract

An anti-magic labeling of a finite simple undirected graph with p vertices and q edges is a bijection from the set of edges to the set of integers $\{1, 2, \dots, q\}$ such that the vertex sums are pairwise distinct, where the vertex sum at vertex u is the sum of labels of all edges incident to such vertex. A graph is called anti-magic if it admits an anti-magic labeling.

Hartsfield and Ringel conjectured in 1990 that all connected graphs except K_2 are anti-magic. Recently, N. Alon et al (Dense graphs are anti-magic, *Journal of Graph Theory*, Volume 47, Issue 4, (2004), pp. 297-309) showed that this conjecture is true for dense graphs, i.e. it is true for p -vertex graphs with minimum degree $\Omega(\log p)$. More recently, D. Hefetz (Anti-magic graphs via the combinatorial nullstellensatz, *Journal of Graph Theory*, Volume 50, Issue 4, (2005), pp. 263-272) proved that in particular for a positive integer k , a graph G with 3^k vertices is anti-magic if it admits a K_3 -factor. Tao-Ming Wang (Toroidal grids are anti-magic, **COCOON 2005**, Lecture Notes in Computer Science (LNCS) **3595** (2005), pp. 671-679) showed that higher dimensional torus graphs are antimagic. However the Hartsfield-Ringel conjecture is still unsettled while the research along this direction is extended.

In this article, some new classes of sparse anti-magic graphs are constructed through Cartesian products.

1 Introduction

All graphs in this paper are finite, simple, undirected, and without loops unless otherwise stated. In 1990, Hartsfield and Ringel[4] introduced the

concepts called anti-magic labeling and anti-magic graphs.

DEFINITION 1.1. For a graph $G = (V, E)$ with p vertices and q edges and without any isolated vertex, an **anti-magic** edge labeling is a bijection $f : E \rightarrow \{1, 2, \dots, q\}$, such that the induced vertex sum $f^+ : V \rightarrow \mathbf{N}$ given by $f^+(u) = \sum \{f(uv) : uv \in E\}$ is injective. A graph is called anti-magic if it admits an anti-magic labeling.

Hartsfield and Ringel showed that paths, cycles, complete graphs K_n ($n \geq 3$) are anti-magic. They conjectured that all connected graphs besides K_2 are anti-magic, which remains unsettled. Recently, Alon et al [1] showed that the last conjecture is true for dense graphs. They showed that all graphs with $n(\geq 4)$ vertices and minimum degree $\Omega(\log n)$ are anti-magic. They also proved that if G is a graph with $n(\geq 4)$ vertices and the maximum degree $\Delta(G) \geq 4n - 2$, then G is anti-magic and all complete partite graphs except K_2 are anti-magic. More recently, D. Hefetz [5] proved that, among others, for $k \in \mathbf{N}$, a graph G with 3^k vertices is anti-magic if it admits a K_3 -factor. Also Tao-Ming Wang [6] showed that the Cartesian products of cycles and regular graphs are anti-magic, and in particular, higher dimensional torus graphs are antimagic. For more conjectures and open problems on anti-magic graphs and related type of graph labeling problems, please see the dynamic survey article of Gallian [3].

In this paper, we will introduce new classes of anti-magic graphs through Cartesian products. Among others, the Cartesian product of cycles with regular graphs, using a different approach from previous ones, are shown to be anti-magic. And the Cartesian product of star graphs with regular graphs, and certain type of anti-prism graphs with regular graphs, are also proved to be anti-magic as corollaries of a more general result.

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2 Preliminaries

In the following, Lemma 2.1 and Lemma 2.2 show that paths and cycles are anti-magic, which are basic facts for later use. Although they have been previously proved[4], we still give proofs here for completion:

Lemma 2.1. *The paths P_{n+1} are anti-magic for $n \geq 2$.*

Proof. Without loss of generality, we assign the path P_{n+1} with the vertex set $V(P_{n+1}) = \{u_0, u_1, \dots, u_{n-1}, u_n\}$ and edge set $E(P_{n+1}) = \{u_i u_{i+2} : i = 0, 1, \dots, n-2\} \cup \{u_{n-1} u_n\}$. We define a bijection (please see Figure 1) $f : E(P_{n+1}) \rightarrow \{1, 2, \dots, n\}$ by

$$\begin{cases} f(u_i u_{i+2}) = i+1 & \text{if } 0 \leq i \leq n-2 \\ f(u_{n-1} u_n) = n \end{cases}$$

And the associated induced vertex sum $f^+ : V(P_{n+1}) \rightarrow \mathbf{N}$ can be calculated as follows :

$$f^+(u_i) = \begin{cases} i+1 & \text{if } i = 0, 1 \\ 2i & \text{if } 2 \leq i \leq n-1 \\ 2i-1 & \text{if } i = n \end{cases}$$

So $f^+(u_0) < f^+(u_1) < \dots < f^+(u_n)$ is a strictly increasing sequence, hence pairwise distinct. $\square$

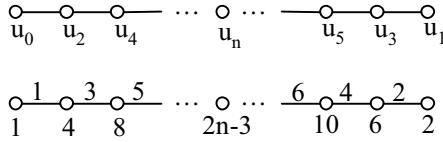


Figure 1: An anti-magic labeling and the associated vertex sums of paths P_{n+1}

Next, we will prove cycles are also anti-magic. We may treat the cycles C_n as the graphs obtained by the paths P_{n+1} through identifying the two end points u_0 and u_1 . Hence an anti-magic labeling of cycle C_n follows from an anti-magic labeling of the path P_{n+1} .

Lemma 2.2. *The cycles C_n are anti-magic for $n \geq 3$.*

Proof. Now, we identify two end vertices (u_0 and u_1) of path P_{n+1} . Without loss of generality, we re-assign the cycle C_n with the vertex set $V(C_n) = \{v_1, v_2, \dots, v_n\}$ and edge set $E(C_n) = \{u_1 u_2\} \cup \{u_i u_{i+2} : i = 1, 2, \dots, n-2\} \cup \{u_{n-1} u_n\}$. We define a bijection (please see Figure 2) $f : E(C_n) \rightarrow \{1, 2, \dots, n\}$ by

$$\begin{cases} f(u_1 u_2) = 1 \\ f(u_i u_{i+2}) = i+1 & \text{if } 1 \leq i \leq n-2 \\ f(u_{n-1} u_n) = n \end{cases}$$

Then the associated induced vertex sum $f^+ : V(C_n) \rightarrow \mathbf{N}$ can be calculated as follows :

$$f^+(u_i) = \begin{cases} 3 & \text{if } i = 1 \\ 2i & \text{if } 2 \leq i \leq n-1 \\ 2i-1 & \text{if } i = n \end{cases}$$

So $f^+(u_1) < f^+(u_2) < \dots < f^+(u_n)$ is strictly increasing, hence pairwise distinct. $\square$

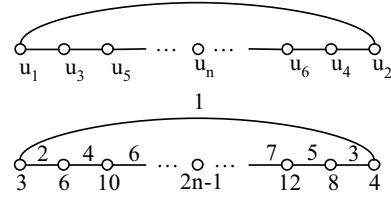


Figure 2: An anti-magic labeling and the associated vertex sums of cycles C_n

3 Cartesian Product of Graphs

In order to obtain an anti-magic labeling of the Cartesian product of graphs, we first consider the translation on an existing anti-magic labeling:

DEFINITION 3.1. *A graph $G = (V, E)$ with $p = |V|$ vertices and $q = |E|$ edges is called **k-anti-magic**, where k is a non-negative integer, if there exists a bijection $f : E \rightarrow \{k+1, k+2, \dots, k+q\}$, such that the induced vertex sum $f^+ : V \rightarrow \mathbf{N}$ given by $f^+(u) = \sum \{f(uv) : uv \in E\}$ is injective. Note that the original anti-magic labeling is 0-anti-magic, in particular.*

However, usually the labeling obtained from the translation on an anti-magic labeling could lose its anti-magic-ness. Therefore, we find the following sufficient condition to have an k -anti-magic translation by k on an existing anti-magic labeling:

Lemma 3.1. *Let G be an anti-magic (p, q) -graph with vertex set $\{v_1, \dots, v_p\}$. Without loss of generality (via renaming vertices), we may assume that the degree sequence is $\deg(v_1) \leq \deg(v_2) \leq \dots \leq \deg(v_p)$. Suppose $f : E(G) \rightarrow \{1, 2, \dots, q\}$ is an anti-magic edge labeling of G with the property that $f^+(v_1) < f^+(v_2) < \dots < f^+(v_p)$, where $f^+ : V(G) \rightarrow \mathbf{N}$ is the induced vertex sum. That is, the ordering of the vertex sums is consistent with*

that of the degree sequence. Let $g : E(G) \rightarrow \mathbf{N}$ be the k -translation of the edge labeling f defined by $g(e) = f(e) + k$, where the integer $k \geq 0$, for each edge e in G , and g^+ is the induced vertex sum from g . Then $g^+(v_1) < g^+(v_2) < \dots < g^+(v_p)$, i.e., the edge labeling g is k -anti-magic.

Proof. $g^+(v_i) = \sum(f(uv_i) + k) = f^+(v_i) + k \cdot \deg(v_i) < f^+(v_{i+1}) + k \cdot \deg(v_{i+1}) = \sum(f(uv_{i+1}) + k) = g^+(v_{i+1})$, for $1 \leq i \leq p-1$. $\square$

If the graphs considered are regular, then the sufficient condition in the above lemma holds trivially. Therefore,

Lemma 3.2. *The following are true:*

1. Suppose the graph G is regular and anti-magic, then it is k -anti-magic, where $k \geq 0$.
2. Suppose G_i is regular and anti-magic for each $1 \leq i \leq n$, then the disjoint union $G_1 + G_2 + \dots + G_n$ is k -anti-magic, where $k \geq 0$. In particular, every 2-regular graph is k -anti-magic, and $nG = G + G + \dots + G$ is k -anti-magic if G is regular and anti-magic.

Proof. For part (1), suppose G is regular and anti-magic, then by the lemma 3.1 the translation by k on an anti-magic labeling of G is k -anti-magic, where the integer $k \geq 0$. As for part (2), let $\deg(G_i)$ be the constant degree of the regular graph G_i . Without loss of generality, we may assume that $\deg(G_1) \leq \deg(G_2) \leq \dots \leq \deg(G_n)$. Also note that from part (1), each G_i is k -anti-magic, hence we may put the edge labels in order to obtain an anti-magic labeling of the disjoint union $G_1 + G_2 + \dots + G_n$. Again by lemma 3.1, the disjoint union is k -anti-magic. $\square$

For convenience, we will use the following convention of notations throughout the paper, unless otherwise stated:

CONVENTION

Given two graph G and H with $|V(G)| = p_1$, $|E(G)| = q_1$ and $|V(H)| = p_2$, $|E(H)| = q_2$, respectively. The Cartesian product of graphs G and H is the graph $G \times H$ with vertex set $V(G) \times V(H)$, and (u_i, v_j) is adjacent to (u_k, v_l) whenever (1) $u_i = u_k$ and $v_j v_l \in E(H)$, or (2) $v_j = v_l$ and $u_i u_k \in E(G)$. Note that $G \times H$ can be decomposed into p_2 isomorphic slices G'_j s of G , together with p_1 isomorphic slices H'_i s of H . Namely, the subgraphs G'_j s are induced from the vertices (u_i, v_j) ,

where j is fixed and i is running from 1 to p_1 , and the subgraphs H'_i s are induced from the vertices (u_i, v_j) , where i is fixed and j is running from 1 to p_2 .

In graphs G and H , without loss of generality, for edges $e_i \in E(G)$ and $\varepsilon_j \in E(H)$ there are bijections $g : E(G) \rightarrow \{1, 2, \dots, q_1\}$ given by $g(e_i) = i$, where $1 \leq i \leq q_1$, and $h : E(H) \rightarrow \{1, 2, \dots, q_2\}$ given by $h(\varepsilon_j) = j$, where $1 \leq j \leq q_2$, such that the induced vertex sum $g^+ : V(G) \rightarrow \mathbf{N}$ has the ordering $g^+(u_1) \leq g^+(u_2) \leq \dots \leq g^+(u_{p_1})$, and the vertex sum $h^+ : V(H) \rightarrow \mathbf{N}$ has the ordering $h^+(v_1) \leq h^+(v_2) \leq \dots \leq h^+(v_{p_2})$, respectively. In the graph $G \times H$, the vertex $(u_i, v_j) \in V(G) \times V(H)$ is represented by $w_{i,j}$, and the edge e_i in the subgraph G_l is represented by $e_{i,l}$, and the edge ε_j in the subgraph H_k is represented by $\varepsilon_{k,j}$, respectively.

Let the bijection $f : E(G \times H) \rightarrow \{1, 2, \dots, p_1 q_2 + p_2 q_1\}$ be an edge labeling of $G \times H$, and the induced vertex sum is f^+ . We denote f_G and f_H to be the edge labeling f restricted to the slices of G_l , $1 \leq l \leq p_2$, and H_k , $1 \leq k \leq p_1$, respectively. And f_G^+ and f_H^+ represent the induced vertex sums respectively. Therefore, we have that $f^+(w_{i,j}) = f_G^+(w_{i,j}) + f_H^+(w_{i,j})$, $1 \leq i \leq p_1$ and $1 \leq j \leq p_2$.

Please see the Figure 3 as an example for the conventions mentioned above.

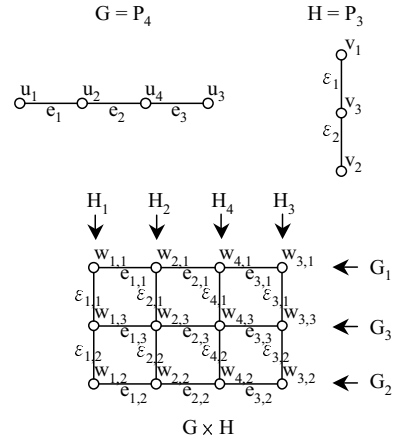


Figure 3: Conventions of notations for Cartesian product $G \times H$

4 Toroidal Grid Graphs Are Antimagic

If G is a d -regular graph ($d \geq 1$), we call $G \times C_n$ ($n \geq 3$) a **generalized toroidal grid graph**.

When G is a cycle, the graph $C_m \times C_n$ is called a **toroidal grid** graph. Note that the antimagicness of toroidal grid graph and generalized toroidal grid graphs have been given in [6], however here we try to present another approach to unify several methods altogether, and will have more corollaries coming along. We show that toroidal grid graphs are antimagic by proving a more general fact as in the following theorem:

Theorem 4.1. *The generalized toroidal grid graphs, i.e. the Cartesian products of cycles and regular graphs, are anti-magic.*

Proof. Let G be the cycles C_n and H be a d -regular graph with $|V(H)| = p$ and $|E(H)| = q$. Since $G = C_n$ is anti-magic, we have the order $g^+(u_1) < g^+(u_2) < \dots < g^+(u_n)$ by using the edge labeling $g(e_i) = i$ for $1 \leq i \leq n$ as in Lemma 2.2. In general, with the edge labeling $h(\varepsilon_i) = i$ for $1 \leq i \leq q$, we have the order $h^+(v_1) \leq h^+(v_2) \leq \dots \leq h^+(v_p)$, and $h^+(v_p) - h^+(v_1) < dq$. Here we still follows the convention on the notations of Cartesian products of graphs mentioned earlier.

Therefore, we label the edges of the slices of H_k and G_l by $f_H(\varepsilon_{k,j}) = (k-1)(p+q) + h(\varepsilon_j)$, $1 \leq k \leq n$, $1 \leq j \leq q$, and $f_G(e_{i,l}) = q + (p+q)(g(e_i) - 1) + l$, $1 \leq i \leq n$, $1 \leq l \leq p$. Hence the vertex sums

of the slices H_k and G_l are $f_H^+(w_{k,j}) = d(k-1)(p+q) + h^+(v_j)$, $1 \leq j \leq q$, $1 \leq k \leq n$, and $f_G^+(w_{i,l}) = 2q + (p+q)(g^+(u_i) - 2) + 2l$, $1 \leq i \leq n$, $1 \leq l \leq p$.

We observe that the vertex sums f_H^+ and f_G^+ have the ordering $f_H^+(w_{i,1}) \leq f_H^+(w_{i,2}) \leq \dots \leq f_H^+(w_{i,p})$ and $f_G^+(w_{i,1}) < f_G^+(w_{i,2}) < \dots < f_G^+(w_{i,p})$, where i is running from 1 to n . Therefore $f^+(w_{i,1}) < f^+(w_{i,2}) < \dots < f^+(w_{i,p})$, where $1 \leq i \leq n$. Hence if we can show $f^+(w_{i,p}) < f^+(w_{i+1,1})$ for $1 \leq i \leq n-1$, then it is anti-magic. But notice that we have $f^+(w_{i,p}) < f^+(w_{i+1,1}) \iff h^+(v_p) - h^+(v_1) < dq + dp + (p+q)(g^+(u_{i+1}) - g^+(u_i)) + 2(1-p)$ for $1 \leq i \leq n-1$, and the latter is true, hence we are done. $\square$

Corollary 4.1.1. *All toroidal grid graphs $C_m \times C_n$ are antimagic for $m, n \geq 3$.*

Proof. Since a toroidal grid graph is the Cartesian product of two cycles and cycles are 2-regular. $\square$

Corollary 4.1.2. *All higher dimensional toroidal grid graphs $C_{m_1} \times C_{m_2} \times \dots \times C_{m_t}$ are antimagic for integer $t \geq 3$.*

Proof. Since the Cartesian product of cycles is regular. $\square$

5 Cartesian Products of Graphs with Regular Graphs

With the above results in mind, we may obtain a more general situation as follows, which can be applied to find anti-magic labeling of the Cartesian product of various types of graphs with regular graphs. We will give examples in the following corollaries.

Theorem 5.1. *Assume G is a graph with $|V(G)| = p_1$ and $|E(G)| = q_1$. Without loss of generality by renaming the vertices, may assume that $g^+(u_1) \leq g^+(u_2) \leq \dots \leq g^+(u_{p_1})$ for an given edge labeling g . Let H be a d -regular graph with $|V(H)| = p_2$ and $|E(H)| = q_2$. If we have the inequalities $p_2 \cdot [g^+(u_{i+1}) - g^+(u_i)] + p_1 \cdot q_2 \cdot [deg(u_{i+1}) - deg(u_i)] \geq (p_2 - 1) \cdot deg(u_{i+1})$ for $i = 1, 2, \dots, p_1 - 1$, then the Cartesian product $G \times H$ is anti-magic.*

Proof. Without loss of generality by renaming the vertices and edges, we may assume that $g^+(u_1) \leq g^+(u_2) \leq \dots \leq g^+(u_{p_1})$ with the edge labeling $g(e_i) = i$ on G , $1 \leq i \leq q_1$, and with the edge labeling $h(\varepsilon_i) = i$ on H , $1 \leq i \leq q_2$, we have the

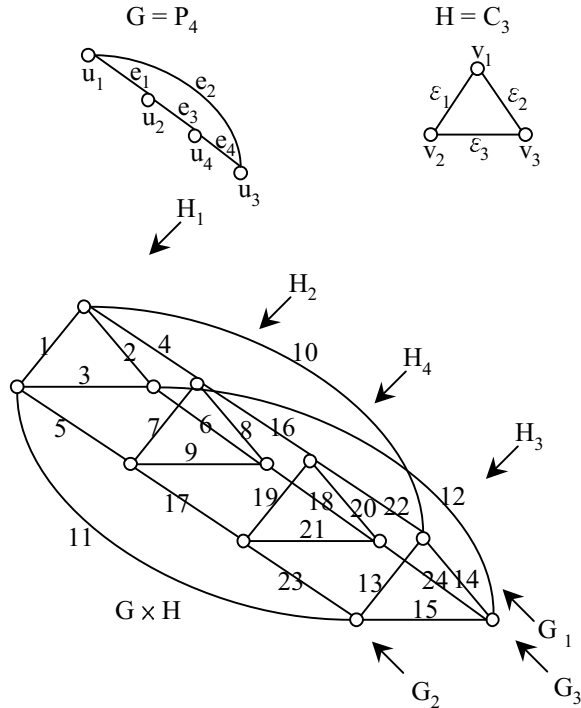


Figure 4: An Anti-magic Labeling of $C_4 \times C_3$

ordering $h^+(v_1) \leq h^+(v_2) \leq \dots \leq h^+(v_{p_2})$, and also $h^+(v_{p_2}) - h^+(v_1) < d \cdot q_2$. The notations here follow the conventions earlier mentioned.

Therefore, we label the edges of the slices of H_k and G_l by $f_H(\varepsilon_{k,j}) = q_2(k-1) + h(\varepsilon_j)$, $1 \leq k \leq p_1$, $1 \leq j \leq q_2$ and $f_G(e_{i,l}) = p_1q_2 + p_2(g(e_i) - 1) + l$, $1 \leq i \leq q_1$, $1 \leq l \leq p_2$. Hence the vertex sums of the slices of H_k and G_l are $f_H^+(w_{k,j}) = dq_2(k-1) + h^+(v_j)$, $1 \leq k \leq p_1$, $1 \leq j \leq q_2$, and $f_G^+(w_{i,l}) = p_2g^+(u_i) + \deg(u_i)(p_1q_2 - p_2 + l)$, $1 \leq i \leq q_1$, $1 \leq l \leq p_2$.

The vertex sums f_H^+ and f_G^+ have the ordering $f_H^+(w_{i,1}) \leq f_H^+(w_{i,2}) \leq \dots \leq f_H^+(w_{i,p_2})$ and $f_G^+(w_{i,1}) < f_G^+(w_{i,2}) < \dots < f_G^+(w_{i,p_2})$, when i ranges from 1 to p_1 . So is $f^+(w_{i,1}) < f^+(w_{i,2}) < \dots < f^+(w_{i,p})$, when i ranges from 1 to p_1 . Hence if we can show $f^+(w_{i,p}) < f^+(w_{i+1,1})$ for $1 \leq i \leq p_1 - 1$, then it is anti-magic. So we observe that $f^+(w_{i,p_2}) < f^+(w_{i+1,1}) \iff h^+(v_{p_2}) - h^+(v_1) < dq_2 + p_2[g^+(u_{i+1}) - g^+(u_i)] + p_1q_2[\deg(u_{i+1}) - \deg(u_i)] - \deg(u_{i+1})(p_2 - 1)$, and the latter inequality is true whenever $p_2[g^+(u_{i+1}) - g^+(u_i)] + p_1q_2[\deg(u_{i+1}) - \deg(u_i)] - \deg(u_{i+1})(p_2 - 1) \geq 0$. Hence we are done. $\square$

Corollary 5.1.1. *Assume G is an anti-magic (p_1, q_1) -graph, and in addition that for the vertices $u_1, \dots, u_{p_1}$ of G , we have consistent degree sequence and vertex sum ordering, i.e. $\deg(u_1) \leq \deg(u_2) \leq \dots \leq \deg(u_{p_1})$ and $g^+(u_1) \leq g^+(u_2) \leq \dots \leq g^+(u_{p_1})$ for an anti-magic edge labeling g . Suppose H is a d -regular (p_2, q_2) -graph. If we have the inequalities*

$$g^+(u_{i+1}) - g^+(u_i) \geq \deg(u_{i+1}),$$

for $i = 1, 2, \dots, p_1 - 1$, then the Cartesian product $G \times H$ is anti-magic.

Proof. Since $\deg(u_{i+1}) - \deg(u_i)$ is non-negative and $g^+(u_{i+1}) - g^+(u_i) \geq \deg(u_{i+1})$ for $i = 1, 2, \dots, p_1 - 1$, the inequalities in the Theorem 5.1 are satisfied. $\square$

With the above mentioned conditions, we may get the anti-magic labeling of various types of graph products. Let us give three examples below. First of all, notice that the star graph $ST(n)$ is the complete bipartite graph $K_{1,n}$.

Corollary 5.1.2. *The Cartesian product of the star graphs $ST(n)$, $n \geq 3$, and regular graphs is anti-magic.*

Proof. Let $ST(n)$ be the star graph with the vertex set $\{u_1, \dots, u_{n+1}\}$, and the vertex u_{n+1} is

the one incident to the other n pendant vertices $u_1, \dots, u_n$. Notice that we have consistent degree sequence and vertex sum orderings as $\deg(u_1) = 1 \leq \dots \leq \deg(u_n) = 1 \leq \deg(u_{n+1}) = n$ and $g^+(u_1) = 1 \leq \dots \leq g^+(u_n) = n \leq g^+(u_{n+1}) = 1 + 2 + \dots + n$ for the anti-magic edge labeling g with $g(u_i u_{n+1}) = i$ for $1 \leq i \leq n$. Hence the above inequalities in Corollary 5.1.1 can be easily checked. $\square$

The double star graph $ST(m, n)$ is a graph that is formed by two stars $ST(m)$ and $ST(n)$ via joining their centers by an edge (please see the Figure5). We give the anti-magic labeling of the Cartesian product of the double star graphs $ST(m, n)$ with regular graphs.

Corollary 5.1.3. *The Cartesian product of the double star graphs $ST(m, n)$, where $m, n \geq 2$, and regular graphs is anti-magic.*

Proof. Without loss of generality, we may assume that $m \leq n$ in the graph $ST(m, n)$. Let u and v be the center vertices of the graph $ST(m)$ and $ST(n)$, respectively, and let $u_1, u_2, \dots, u_m$ be the neighbors of u , and $v_1, v_2, \dots, v_n$ be the neighbors of v . Then we assign $1, 2, \dots, m$ to the edges $uu_1, uu_2, \dots, uu_m$, and assign $m+1, m+2, \dots, m+n$ to the edges $vv_1, vv_2, \dots, vv_n$, and assign $m+n+1$ to the edge uv . Then we check whether it satisfies the sufficient conditions in Corollary 5.1.1. It

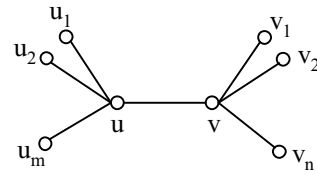


Figure 5: Double Star Graph $ST(m, n)$

is clear to see that over the vertices with degree 1 the inequalities are satisfied. So we only need to check two inequalities $f^+(u) - f^+(v_n) \geq \deg(u)$ and $f^+(v) - f^+(u) \geq \deg(v)$. For the first one, $f^+(u) - f^+(v_n) = \frac{m(m+1)}{2} + 1$, it must be greater than $\deg(u) = m + 1$. For the second inequality, $f^+(v) - f^+(u) = \frac{2mn + (n^2 + n - m^2 - m)}{2} \geq mn$ is greater than $n + 1$ if $n \geq m \geq 2$, and $f^+(v) - f^+(u) = \frac{2n + (n^2 + n - 2)}{2}$ is greater than $n + 1$ if $n > m = 1$. So we are done. $\square$

An anti-prism with $2n$ vertices ($n \geq 3$) is the polyhedron composed of two parallel copies of some particular n -sided polygon, connected by an

alternating band of triangles. The graph corresponding to the skeleton of an anti-prism is known, not surprisingly, as an anti-prism graph. We have the following:

Corollary 5.1.4. *Let G be anti-prism with $2n$ vertices for $n \geq 3$, $n \not\equiv 2 \pmod{4}$. Then the Cartesian product of graphs G and regular graphs is anti-magic.*

Proof. Since G is a regular graph of degree 4, and it is $(4n+4, 4)$ -antimagic [2], where a graph is called (a, d) -antimagic if it is antimagic and also the vertex sums form an arithmetic sequence $a, a+d, \dots, a+(|V|-1)d$. The conditions in the Theorem 5.1 or in the Corollary 5.1.1 can be checked easily and applied in this case. $\square$

6 Conclusion and Future Studies

While we obtained the above mentioned results, the antimagic labeling of certain class of trees are also found. However this is not within the same scope of this article, and we will report in subsequent papers. Definitely it will be very interesting if one can show that all trees except K_2 are antimagic, before the Hartsfield-Ringel conjecture is settled.

On the other hand, inspired from our results here, a more general situation regarding the antimagic-ness of regular graphs is unknown and is interesting to explore. Many special cases of regular graphs have been investigated, and it is evident that we need the last piece of the jigsaw puzzle.

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